# SVOJSTVA FOURIEROVE TRANSF

## Linearnost

$$\text{DTFT}\{a f[n] + b g[n]\} = a F(e^{j\Omega}) + b G(e^{j\Omega})$$

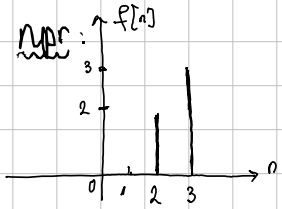
isto vrijedi i za ostale transf. pa ne moramo svd pisati

periodičnost

$$= \sum_{n=-\infty}^{\infty} \left(a f[n] + b g[n]\right) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} a f[n] e^{-j\Omega n} + \sum_{n=-\infty}^{\infty} b g[n] e^{-j\Omega n}$$

$$= a \underbrace{\sum_{n=-\infty}^{\infty} f[n] e^{-j\Omega n}}_{\text{F. transf}} + b \underbrace{\sum_{n=-\infty}^{\infty} g[n] e^{-j\Omega n}}_{\text{F. transf}} = a F(e^{j\Omega}) + b F(e^{j\Omega})$$

npr:




$2\delta[n-2] + 3\delta[n-3]$

pomaknuti impulsi

$\longrightarrow 2e^{-j2\Omega} + 3e^{-j3\Omega}$

Kada je broj realan ali negativan, pojavljuje se faza $\pi$

## Konjugirana simetričnost

vrijedi samo za realne signale

$$\text{CTFT}: \quad F(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$F^*(j\omega) \rightarrow$ svugdje gdje je $j$ treba staviti $-j$ (konjugirano kompleksni spektar)

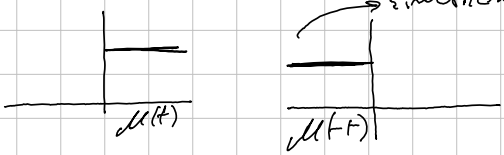
$$F^*(j\omega) = \int_{-\infty}^{\infty} f^*(t) e^{j\omega t} dt$$

$$f(t) = f^*(t)$$

dobili smo spektar od MINUS OMEGE

$$F^*(j\omega) = \int_{-\infty}^{\infty} f^*(t) e^{-j(-\omega t)} dt$$

$$F^*(-j\omega) = \int_{-\infty}^{\infty} f^*(t) e^{-j(-\omega)t} dt$$




$$\text{Re}\{F(j\omega)\} = \text{Re}\{F(-j\omega)\} \qquad |F(j\omega)| = |F(-j\omega)|$$

$$\longrightarrow$$

$$\text{Im}\{F(j\omega)\} = -\text{Im}\{F(-j\omega)\} \qquad \measuredangle F(j\omega) = -\measuredangle F(-j\omega)$$

## Svojstvo pomaka u vremenskoj domeni

$$f[n] \xrightarrow{DTFT} F(e^{j\Omega})$$

$$f[n-m] \longrightarrow |F(e^{j\Omega})|\, e^{-j\Omega m}$$ pomak u fazi

supstitucija

sume radimo po r

$$F_1[e^{j\Omega}] = \sum_{n=-\infty}^{\infty} f[n-m]\, e^{-j\Omega n} = \left\{ \begin{array}{l} n-m=r \\ n=r+m \end{array} \right\} = \sum_{r=-\infty}^{\infty} f[r]\, e^{-j(r+m)}$$

$$= \sum_{r=-\infty}^{\infty} f[r]\cdot e^{-jr} \cdot \underbrace{e^{-jm}}_{konst} = e^{-jm} \cdot \underbrace{\sum_{n=-\infty}^{\infty} f[r]\cdot e^{-jr}}_{def.\ F.\ transf} \Longrightarrow e^{-j\Omega m} \cdot F(e^{j\Omega})$$

$$f[n] \xrightarrow{DTFT} F(e^{j\Omega})$$

$$e^{j\Omega_1 n} f[n] \xleftarrow{IDTFT} F(e^{j(\Omega-\Omega_1)})$$

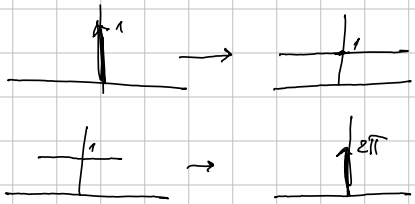
**Svojstvo dualnosti** spomenuli smo kod CTFT (lagano, u frekv. i vremenskoj imamo konf. spektar)

DTFT → kod nekih imamo kontinuiranost

kod nekih ne iako su dualne

$$f(t) \xrightarrow{CTFT} F(j\omega)$$

$$F(jt) \xrightarrow{ICTFT} 2\pi \cdot f(-\omega)$$




$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)\, e^{j\omega t}\, d\omega$$

uvedimo pomoćnu varijablu $\tau \rightarrow t = -\tau$

$$f(-\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)\cdot e^{-j\omega\tau}\, d\omega \qquad \omega = t$$

$$f(-\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(jt)\, e^{-jt\tau}\, dt \qquad \tau = \omega$$

$$f(-\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(jt)\, e^{-jt\omega}\, dt \longrightarrow 2\pi f(-\omega) = CTFT(F(jt))$$

# Konvolucija → FT = aperiodičnost

→ određujemo konvoluciju dva aperiodična vremenski diskretna signala $f[n]$ i $g[n]$

$$f[n] * g[n] = \sum_{m=-\infty}^{\infty} f[m] \cdot g[n-m]$$

$$DTFT\{f[n] * g[n]\} = F(e^{j\Omega}) \cdot G(e^{j\Omega})$$

• za periodične vremenski kont. signale $f$ i $g$ omovre periode $T_0$

$$(f \circledast g)(t) = \int_0^{T_0} f(\tau) g(t-\tau) d\tau$$

→ u slučaju da su periodični signali – Fourierova sumacija

↳ moramo pomnožiti s periodom jer im periodi moraju biti isti

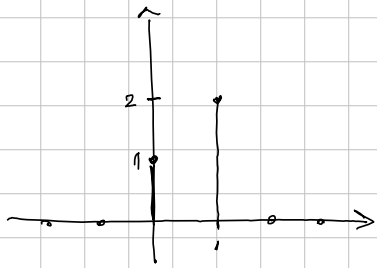
$$(f \circledast g)[n] \xleftarrow{DTFS:\ \Omega = \frac{2\pi}{N}} N F_k \cdot G_k$$

$$(f \circledast g)(t) \xleftarrow{CTFS:\ \omega_0 = \frac{2\pi}{T_0}} T_0 \cdot F_k \cdot G_k$$

Primjer:

$$f[n] = \{1, 2\} = 1 \cdot \delta[n] + 2 \cdot \delta[n-1]$$

$$g[n] = \{-1, 1\} = -1 \cdot \delta[n] + \delta[n-1]$$




(1) prebacimo u frekv. domenu

$$F(e^{j\Omega}) = 1 + 2e^{-j\Omega}$$

$$G(e^{j\Omega}) = -1 + e^{-j\Omega}$$

(2) pomnožimo

konv:

$$\longrightarrow F(e^{j\Omega}) \cdot G(e^{j\Omega})$$

$$= (1 + 2e^{-j\Omega})(-1 + e^{-j\Omega})$$

$$= -1 + e^{-j\Omega} - 2e^{-j\Omega} + 2e^{-2j\Omega}$$

$$= -1 - e^{-j\Omega} + 2e^{-2j\Omega}$$

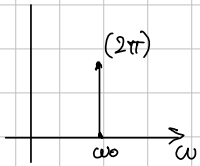
(3) vratimo u vremensku:

$$\boxed{h[n] = -\delta[n] - \delta[n-1] + 2\delta[n-2]}$$

$$f[n]\cdot g[n] \xrightarrow{DTFT} \frac{1}{2\pi} F(e^{j\Omega}) \otimes G(e^{j\Omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(e^{j\psi})\cdot G(e^{j(\Omega-\psi)})\,d\psi$$

Primjer: $F(j\omega) = 2\pi\delta(\omega-\omega_0)$

CTFT



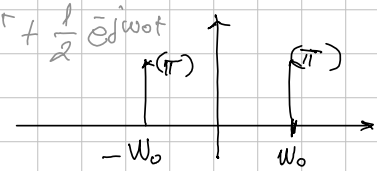

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)\, e^{j\omega t}\, dt$$

$$= \frac{1}{\cancel{2\pi}}\int_{-\infty}^{\infty} \cancel{2\pi}\delta(\omega-\omega_0)\, e^{j\omega t}\, dt = e^{j\omega_0 t}$$

$\omega=\omega_0$

$$= \cos\omega_0 t + j\sin\omega_0 t$$

$$* \cos\omega_0 t = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$




$$f(t) = \cos\omega_0 t = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

$$F(j\omega) = \frac{1}{\cancel{2}}\cdot 2\pi\delta(\omega-\omega_0) + \frac{1}{\cancel{2}}\cancel{2}\pi(\omega+\omega_0)$$

$$= \pi\delta(\omega-\omega_0) + \pi(\omega+\omega_0)$$

- umnožak kontinuiranog signala s pomaknutim impulsom, ne mijenjamo signal već ga samo ponišćuje

Primjer:

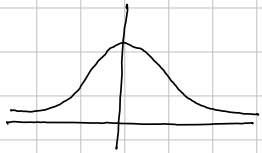
$$DTFT\{f[n]\cdot g[n]\} = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\pi\delta(\psi+\Omega_0) + \pi\delta(\psi-\Omega_0)\right)\cdot G(e^{j(\Omega-\psi)})\,d\psi$$

$$= \frac{1}{\cancel{2\pi}}\int_{-\pi}^{\pi}\cancel{\pi}\delta(\psi+\Omega_0)\, G(e^{j(\Omega-\psi)})\,d\psi + \frac{1}{\cancel{2\pi}}\int_{-\pi}^{\pi}\cancel{\pi}\delta(\psi-\Omega_0)\cdot G(e^{j(\Omega-\psi)})\,d\psi$$

$\psi=-\Omega_0$ $\qquad$ $\psi=\Omega_0$

$$= \frac{1}{2}G(e^{j(\Omega+\Omega_0)}) + \frac{1}{2}G(e^{j(\Omega-\Omega_0)})$$

Gaussov puls:



vremenska:

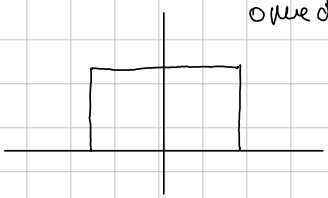
frekv.

→ i u vremenskoj i
u frekv. je neomeđeno (nikad ne ode u 0)

▸ više primjera:

vremenska domena: frekvencijska domena:

omeđen.



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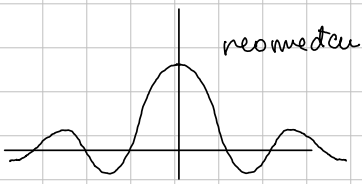

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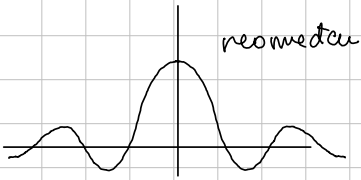

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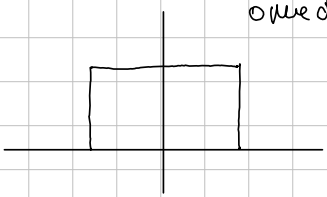

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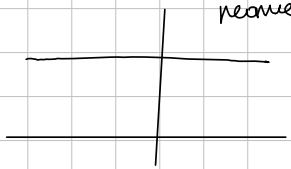

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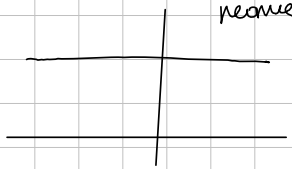
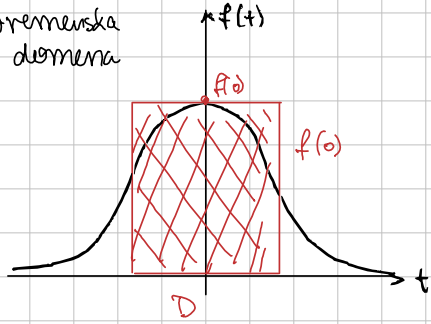


omeđen

ne postoji konk. da je omeđen u obje domene

# Gaussov puls:

vremenska domena

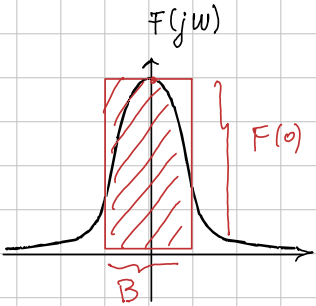



Površinu inače računamo kao integral od tg područja:

$$\int_{-\infty}^{\infty} f(t)\,dt = f(0)\cdot D$$

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)\,e^{-j\omega t}\,dt$$

$$F(0) = \int_{-\infty}^{\infty} f(t)\,e^{-j\cdot 0\cdot t}\,dt = \underline{\underline{f[0]\cdot D}}$$




$$\int_{-\infty}^{\infty} F(j\omega)\,d\omega = F(0)\cdot B$$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)\cdot e^{j\omega t}\,d\omega$$

$$f(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)\,e^{j\omega\cdot 0}\,d\omega = \underline{\underline{\frac{1}{2\pi}F(0)\cdot B}}$$

$$\Rightarrow D = \frac{F(0)}{f(0)} \qquad B = \frac{f(0)}{F(0)}\cdot 2\pi$$

$$D = \frac{2\pi}{B} \longrightarrow \boxed{D\cdot B = 2\pi}$$

Što je šire u drugoj, bit će uže u drugoj

$\Rightarrow$ obrnuto proporcionalno

$\Rightarrow$ zato nema signala koji su omeđeni u obje domene

$\Rightarrow$ Heisenberova neodređenost

* nešto za kraj ne slušam više

$$e^{j\omega_0 t} \longrightarrow 2\pi\,\delta(\omega-\omega_0)$$

$$\cos\omega_0 t = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t} \longrightarrow \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$$

$$f(t) = \sum_{k=-\infty}^{\infty} F_k e^{jk\omega_0 t} \qquad \text{ICTFS}$$

$$F(j\omega) = \sum_{k=-\infty}^{\infty} F_k\cdot 2\pi\,\delta(\omega-k\omega_0)$$